

Problem Set 8: Solutions

SPEA V706 Math Camp | Module 8 · Wednesday, August 12

Part A. Concepts

A1. Moments from a mass function

a. Every probability is between 0 and 1, and they sum to $0.40 + 0.30 + 0.20 + 0.07 + 0.03 = 1.00$. Valid.

b. Weight each value by its probability:

$$\begin{aligned} E[X] &= 0(.40) + 1(.30) + 2(.20) + 3(.07) + 4(.03) \\ &= 0 + .30 + .40 + .21 + .12 = 1.03 \end{aligned}$$

c. Same weights, applied to the squares 0, 1, 4, 9, 16:

$$\begin{aligned} E[X^2] &= 0(.40) + 1(.30) + 4(.20) + 9(.07) + 16(.03) \\ &= 0 + .30 + .80 + .63 + .48 = 2.21 \end{aligned}$$

For contrast, $(E[X])^2 = 1.03^2 = 1.0609$, a very different number. The gap between them is the variance, which is Jensen's inequality showing up early.

d. By V1:

$$\text{Var}(X) = 2.21 - 1.0609 = 1.1491, \quad \sigma_X = \sqrt{1.1491} \approx 1.072$$

e. The average person makes about 1.03 doctor visits per year. A typical person sits about 1.07 visits away from that average, which is a large spread relative to the mean: most people make no visits at all, and a small group makes several.

A2. Linearity in action

a. $S = 50 + 120X$. Split the sum with E3, pull the constant out with E2, and use E1 on the premium:

$$\begin{aligned}
E[S] &= E[50] + E[120X] & (E3) \\
&= 50 + 120 E[X] & (E1 \text{ and } E2) \\
&= 50 + 120(1.03) = \$173.60
\end{aligned}$$

b. By V2, with $a = 120$ and $b = 50$:

$$\text{Var}(S) = 120^2 \text{Var}(X) = 14,400(1.1491) = 16,547.04$$

The \$50 premium drops out because everyone pays it. Adding the same constant to every person shifts the whole distribution sideways without changing how spread out it is.

c. $\sigma_S = \sqrt{16,547.04} \approx \128.64 , and $120 \times 1.072 = 128.64$. The standard deviation scales by $|a|$ while the variance scales by a^2 , which is why σ is the friendlier of the two.

A3. Changing units

a. By linearity, $E[F] = 32 + 1.8(20) = 68^\circ\text{F}$.

b. By V2 the additive 32 drops out and the 1.8 comes out squared:

$$\text{Var}(F) = 1.8^2(25) = 81, \quad \sigma_F = \sqrt{81} = 9^\circ\text{F}$$

c. The days did not change; only the ruler did. A Fahrenheit degree is a smaller unit than a Celsius degree, so the same physical spread takes more of them, 9 instead of 5. Comparing standard deviations across different units is meaningless. If you want a unit-free comparison, use something like the coefficient of variation, or convert both to the same scale first.

A4. Jensen's inequality with two wages

a. $E[W] = \frac{1}{2}(10) + \frac{1}{2}(40) = \25 , so $\log E[W] = \log 25 \approx 3.2189$.

b. Take logs first, then average:

$$E[\log W] = \frac{1}{2} \log 10 + \frac{1}{2} \log 40 \approx \frac{1}{2}(2.3026) + \frac{1}{2}(3.6889) = 2.9957$$

That is exactly $\log 20$, since averaging logs is taking a log of the *geometric* mean, and $\sqrt{10 \times 40} = 20$.

c. $\log E[W] = 3.2189$ is larger than $E[\log W] = 2.9957$. The log is **concave**, so (E5) flips and gives $E[\log W] \leq \log E[W]$. This is the direction predicted, and the gap here is about 0.223.

d. No. A change of 0.10 in mean log wages is approximately a 10% change in the *geometric* mean, not in the average wage. The two coincide only when the wage distribution is degenerate, and the gap grows with the spread of wages. Read “log points” as log points.

A5. A one-line proof

$E[X]$ is a number, not a random variable, so E1 applies to it:

$$\begin{aligned} E[X - E[X]] &= E[X] - E[E[X]] && \text{(E3, the sum splits)} \\ &= E[X] - E[X] && \text{(E1, the expectation of a constant)} \\ &= 0 \end{aligned}$$

In words: the mean is the balance point of the distribution. Deviations above it and below it are equal in total weight, so they cancel exactly. This is also why we square the deviations to build the variance. Averaging them raw always gives zero and tells you nothing about spread.

Part B. R

```
pacman::p_load(data.table, wooldridge)
data(wage1, package = "wooldridge")
dt <- as.data.table(wage1)
```

B1. Look before you leap

```
dim(dt)
```

```
[1] 526 24
```

```
dt[, .N]
```

```
[1] 526
```

```
sum(is.na(dt$wage))
```

```
[1] 0
```

a. 526 rows and 24 columns.

b. `str()` shows the 0/1 indicators: `female`, `married`, `nonwhite`, `smsa`, `south`, `west`, `northcen`, and the industry and occupation dummies (`construc`, `trade`, `services`, `profocc`, and so on). A quick tell is that `summary()` reports a min of 0, a max of 1, and a mean between them, which for a dummy is the proportion.

c. No missing wages. Not every dataset is this cooperative.

B2. The i, j, by model

```
dt[educ > 12, .N] # i only
```

```
[1] 212
```

```
dt[, .(mean_wage = mean(wage), sd_wage = sd(wage))] # j only
```

```
      mean_wage  sd_wage  
      <num>      <num>  
1:    5.896103  3.693086
```

```
dt[, .(mean_wage = mean(wage)), by = female] # j and by
```

```
  female mean_wage  
  <int>      <num>  
1:      1  4.587659  
2:      0  7.099489
```

```
dt[married == 1, .(mean_wage = mean(wage)), by = female] # all three
```

```
  female mean_wage  
  <int>      <num>  
1:      1  4.565909  
2:      0  7.983032
```

- a. 212 workers have more than 12 years of schooling.
- b. Mean wage \$5.90 per hour, standard deviation \$3.69 (1976 dollars).
- c. Women average \$4.59, men \$7.10.
- d. Among married workers the gap is wider: \$4.57 for women, \$7.98 for men. Filtering in *i* before summarizing in *j* is the whole point of the three-slot design.

B3. Creating columns with :=

```
dt[, lwage_check := log(wage)]  
dt[, all.equal(lwage, lwage_check)]
```

```
[1] "Mean relative difference: 2.276759e-08"
```

```
dt[, hs_plus := fifelse(educ >= 12, 1L, 0L)]
dt[, mean(hs_plus)]
```

```
[1] 0.7794677
```

a, b. The check returns TRUE, so `lwage` in the shipped data is exactly `log(wage)`.

c. About 78% of the sample has at least 12 years of schooling. `fifelse()` is the fast, type-safe `data.table` version of `ifelse()`; it insists both branches return the same type, which catches a whole class of silent bugs.

d. `:=` returns the table invisibly, because its job is the *side effect* of modifying `dt` in place. `mean(wage)` in the `j` slot returns a **new** value and leaves `dt` alone. “By reference” means `:=` does not copy the table to change it, which is why `data.table` stays fast on large data. The trap is that a function receiving `dt` and using `:=` modifies the caller’s table too. Use `copy()` when you want a table you can scribble on safely.

B4. Jensen, numerically

```
dt[, mean(log(wage))]
```

```
[1] 1.623268
```

```
dt[, log(mean(wage))]
```

```
[1] 1.774292
```

```
dt[, log(mean(wage)) - mean(log(wage))]
```

```
[1] 0.1510231
```

```
dt[, exp(mean(log(wage)))]
```

```
[1] 5.069633
```

a, b. $E[\log W] \approx 1.6233$ and $\log E[W] \approx 1.7743$. The log of the mean is larger, matching A4 and matching (E5) for a concave function.

c. The gap is about 0.151 log points. `exp(mean(log(wage)))` is \$5.07, while the actual mean wage is \$5.90. Exponentiating an average of logs recovers the **geometric** mean, which always sits below the arithmetic mean when the data vary. That is Jensen’s inequality, in dollars.

B5. Group summaries and `.SDcols`

```
dt[, lapply(.SD, mean), .SDcols = c("wage", "educ", "exper")]
```

	wage	educ	exper
	<num>	<num>	<num>
1:	5.896103	12.56274	17.01711

```
dt[, lapply(.SD, mean), .SDcols = c("wage", "educ", "exper"), by = female]
```

	female	wage	educ	exper
	<int>	<num>	<num>	<num>
1:	1	4.587659	12.31746	16.42857
2:	0	7.099489	12.78832	17.55839

```
dt[, .(mean_wage = mean(wage), sd_wage = sd(wage), n = .N),
  by = .(female, married)][order(-mean_wage)]
```

	female	married	mean_wage	sd_wage	n
	<int>	<int>	<num>	<num>	<int>
1:	0	1	7.983032	4.405032	188
2:	0	0	5.168023	2.722846	86
3:	1	0	4.611583	2.995566	120
4:	1	1	4.565909	2.026258	132

.SD is the sub-table of the columns named in .SDcols, restricted to the current group. lapply(.SD, mean) therefore applies mean down each of those columns without writing the name three times. Chaining a second [] onto the result sorts it, and order(-mean_wage) sorts descending.

B6. Connect it back

a. Married men are highest at \$7.98 per hour; married women are lowest at \$4.57. The gap is about \$3.42 per hour, which is roughly 75% of the married-women mean. Note the ordering: for men, being married goes with much higher wages (\$7.98 versus \$5.17), while for women it goes with slightly lower wages (\$4.57 versus \$4.61).

b. “Among married men in this sample, the average hourly wage is \$7.98.” That number is an estimate of $E[\text{wage} \mid \text{female} = 0, \text{married} = 1]$. You are estimating a conditional expectation function, evaluated at one cell. In two weeks you will call this a saturated regression.

c. The group sizes are 188 married men, 132 married women, 120 single women, and 86 single men. The larger the group, the more we should trust its mean, because averages of more draws bounce around less. On Day 4 that intuition gets a formula: the sampling variance of a mean is σ^2/n , so the standard error of the married-men mean is roughly $\sqrt{188/86} \approx 1.5$ times smaller than that of the single-men mean. A difference in wages between two groups is not evidence of anything until you know how much each mean would wobble across samples.

Check yourself.

1. Because $\text{Var}(X) = E[(X - E[X])^2]$ is an average of squares, and squares are never negative. Equivalently, $E[X^2] \geq (E[X])^2$ is Jensen's inequality applied to the convex function $g(x) = x^2$.
2. Adding a constant b shifts the mean to $E[X] + b$ and leaves the variance untouched. Shifting a distribution does not change its spread.
3. `educ > 12` in the `i` slot filters to the rows with more than 12 years of schooling. `mean(wage)` in the `j` slot computes the average wage among the rows that survived. `by = female` in the `by` slot repeats the `j` computation separately for women and for men, returning one row per group.